

Lesson Objective

Express the length of an object using centimeter cubes as length units to measure with no gaps or overlaps.

Materials: Centimeter cubes, new crayon, dry erase marker, small glue stick

1. Measure the length of a crayon using centimeter cubes. How long is the crayon? <i>Answer: The crayon is 9 centimeter cubes long.</i>	Place a new crayon in front of the student and a pile of centimeter cubes nearby. Model lining up the first centimeter cube so that its edge matches the endpoint of the crayon. Ask the student why it is important to start at the endpoint. Place a second cube end to end with the first, then pause and place a third cube with a small gap. Ask the student what is wrong and have them fix it. Have the student finish lining up centimeter cubes along the crayon with no gaps and no overlaps until the cubes reach the other end. Count the cubes together, touching each one. Ask the student to say the length using the sentence frame: "The crayon is ____ centimeter cubes long."
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2. Choose the picture that shows the correct way to measure the dry erase marker. How long is the marker? Picture A: A dry erase marker with 5 centimeter cubes lined up below it. The first cube starts past the left end of the marker, and the last cube ends before the right end of the marker. There are no gaps between cubes. Picture B: A dry erase marker with 7 centimeter cubes lined up below it. The first cube is lined up with the left end of the marker, but two of the cubes in the middle overlap each other. Picture C: A dry erase marker with 6 centimeter cubes lined up below it. The first cube is lined up with the left end of the marker, and each cube is placed end to end with no gaps and no overlaps, reaching the other end of the marker. <i>Answer: Picture C; the marker is 6 centimeter cubes long.</i>	Read the problem with the student and orient them to the three pictures. Ask the student to look at Picture A and check whether the first cube lines up with the endpoint of the marker. Have them explain why this picture does not show correct measuring. Ask the student to look at Picture B and check for gaps and overlaps. Have them explain what is wrong. Ask the student to look at Picture C and confirm that the cubes start at the endpoint, are placed end to end, and have no gaps or overlaps. Have the student count the cubes and state the length using the sentence frame: "The marker is ____ centimeter cubes long."
3. Measure the length of the glue stick using centimeter cubes. How long is the glue stick?	Place a small glue stick and a pile of centimeter cubes in front of the student. Students work independently.

Answer: The glue stick is 8 centimeter cubes long.

Monitor For

- Does the student line up the first centimeter cube so its edge matches the endpoint of the object?
- Does the student place cubes end to end with no gaps or overlaps as they measure?
- Does the student count each cube one time, touching each cube as they count?
- Does the student state the length using the full sentence frame "The ____ is ____ centimeter cubes long"?

Instructional Moves

- Before placing any cubes in Problem 1, point to the endpoint of the crayon and ask, "Where should the first cube start?" to anchor endpoint alignment.
- If a student leaves gaps between cubes, slide the cubes together and ask, "What happens to the part of the crayon under the gap if we count it that way?"
- If a student overlaps cubes, separate them and ask, "If two cubes cover the same spot, are we counting that part once or twice?"
- If a student loses track while counting, have them touch each cube as they say the number aloud, then restate the length using "The ____ is ____ centimeter cubes long."

Reflection Questions

- Why do we have to line up the first centimeter cube with the end of the object? *Answer: So the cubes start where the object starts and we measure the whole length.*
- What goes wrong when there is a gap between centimeter cubes? *Answer: Some of the object's length is not counted, so the number is too small.*

- What goes wrong when centimeter cubes overlap? *Answer: Part of the object gets counted twice, so the number is too big.*

- How did you know the glue stick was 8 centimeter cubes long? *Answer: I lined up cubes from one end to the other with no gaps or overlaps and counted 8 cubes.*